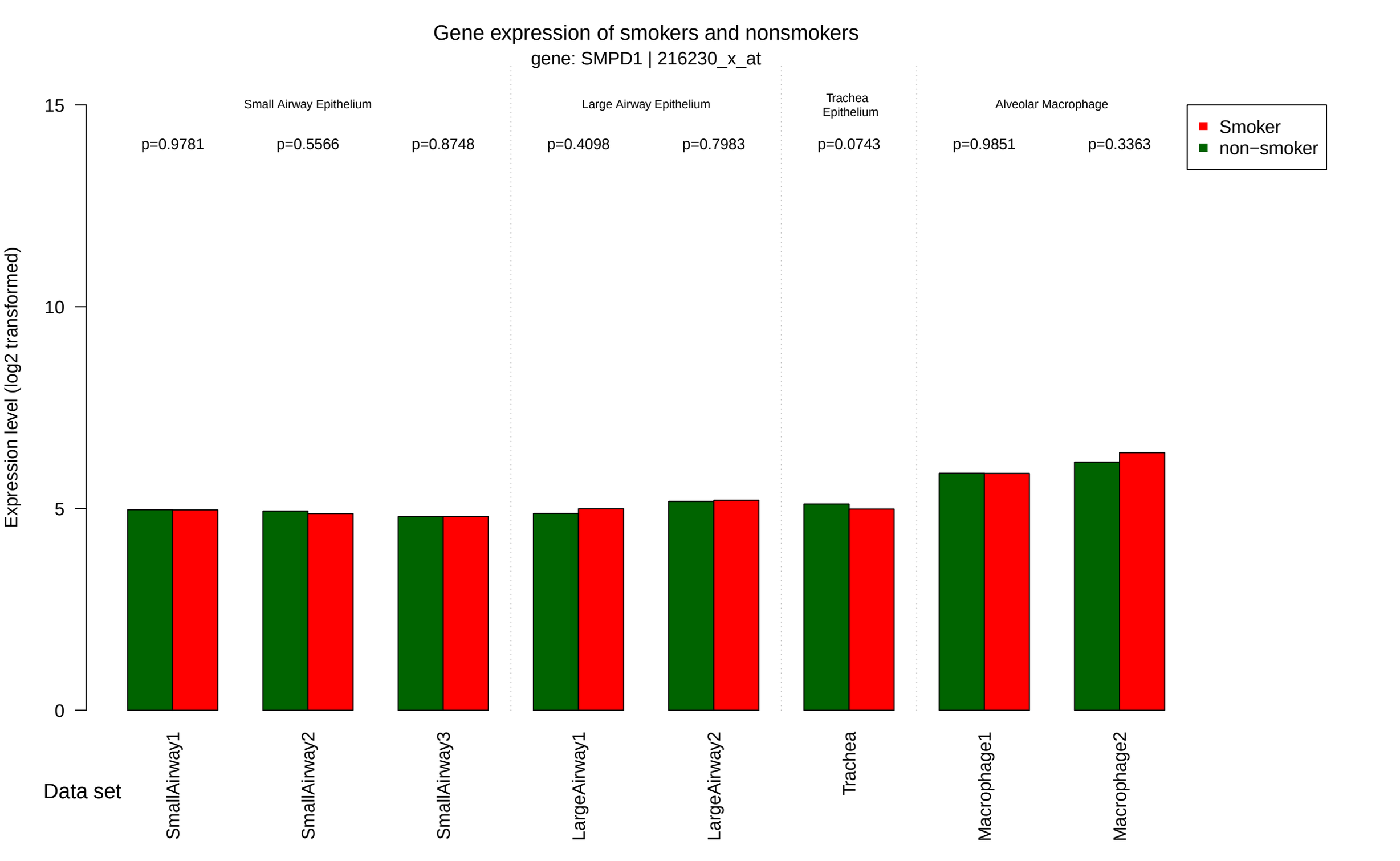
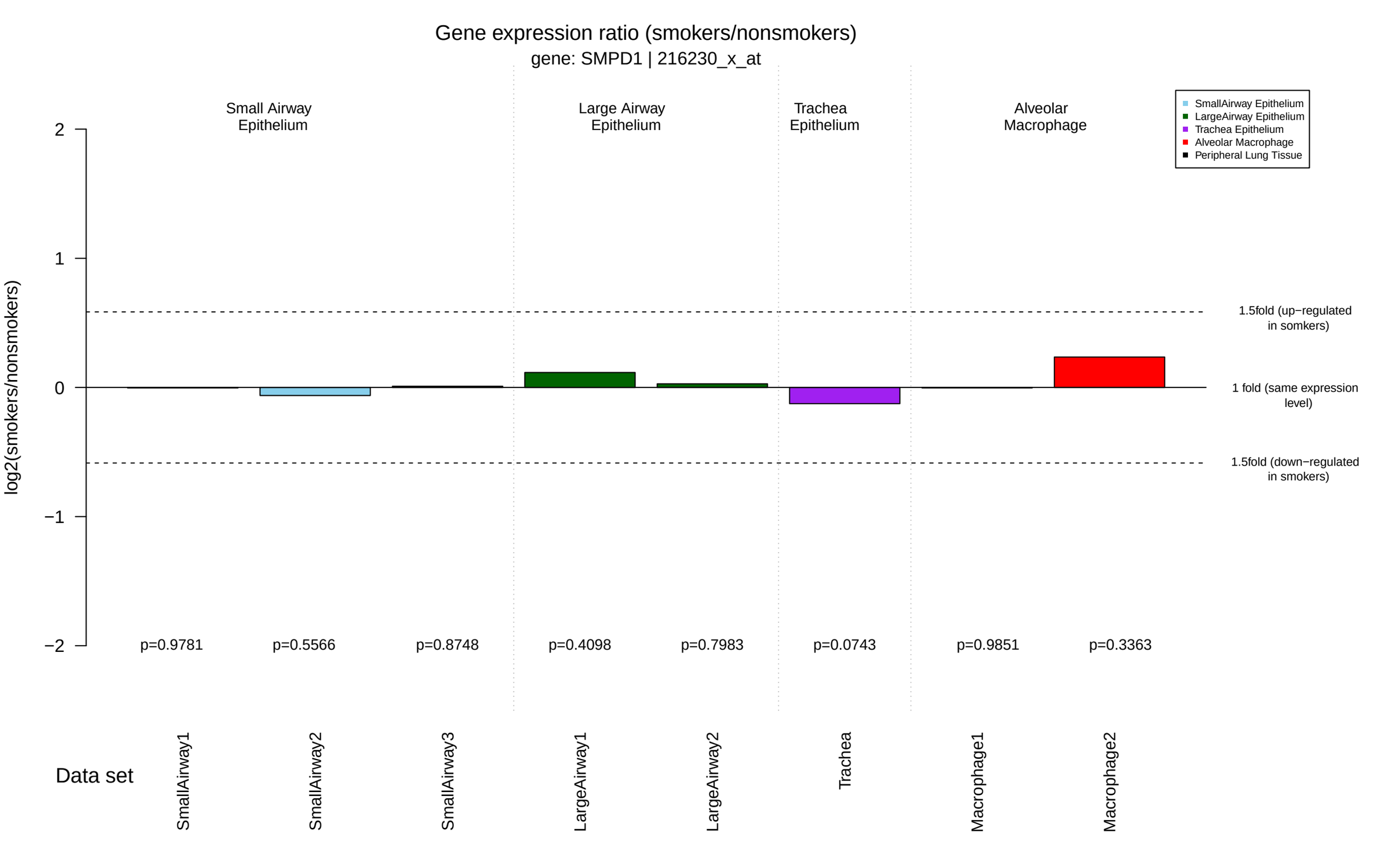
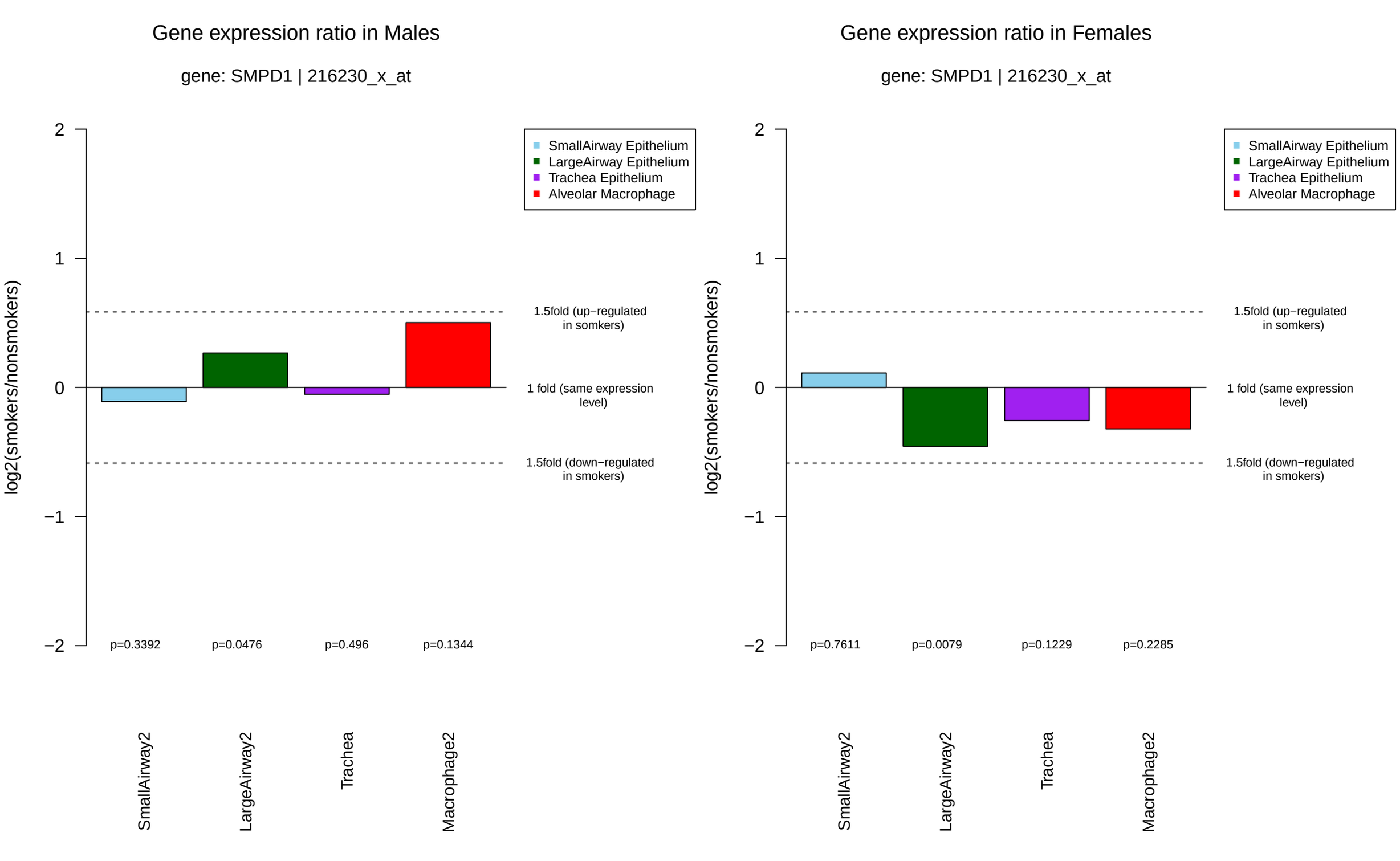
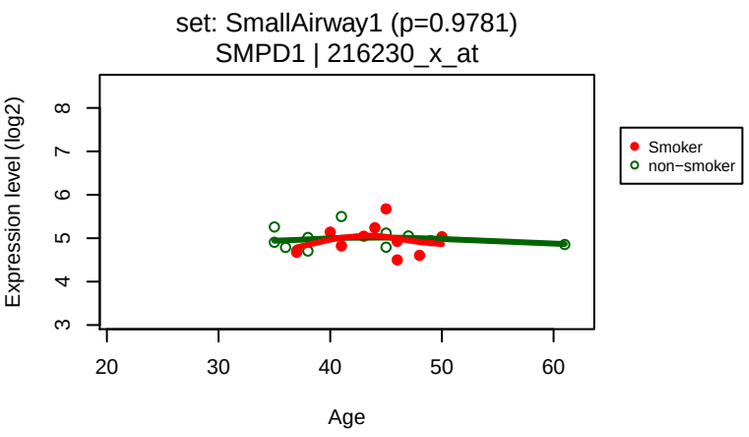
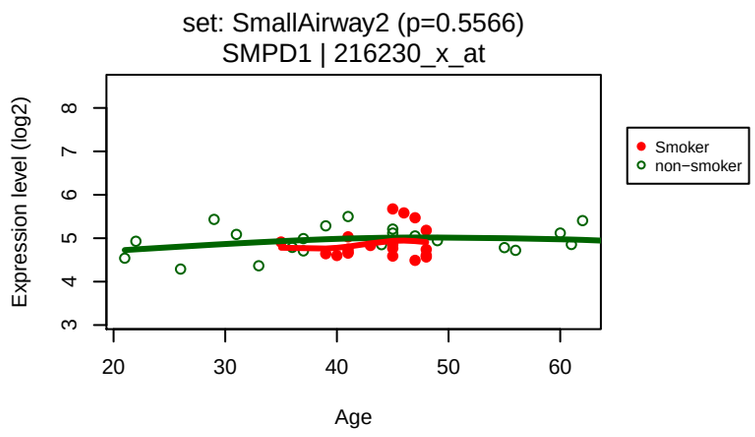
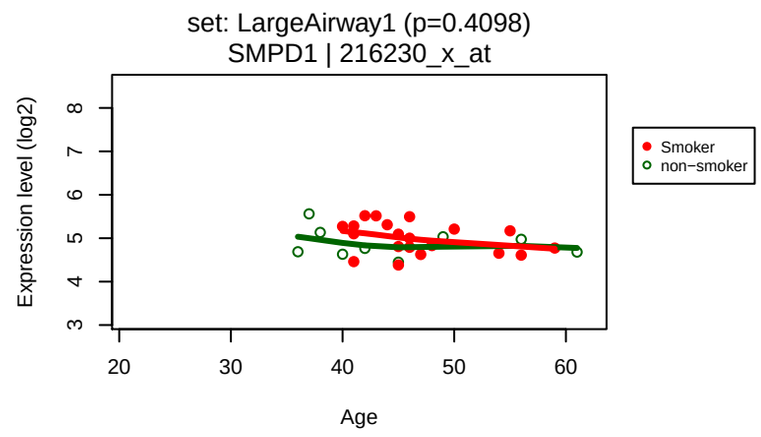

Gene expression of smokers and nonsmokers
gene: SMPD1 | 216230_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9781
p=0.5566
p=0.8748
p=0.4098
p=0.7983
p=0.0743
p=0.9851
p=0.3363
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

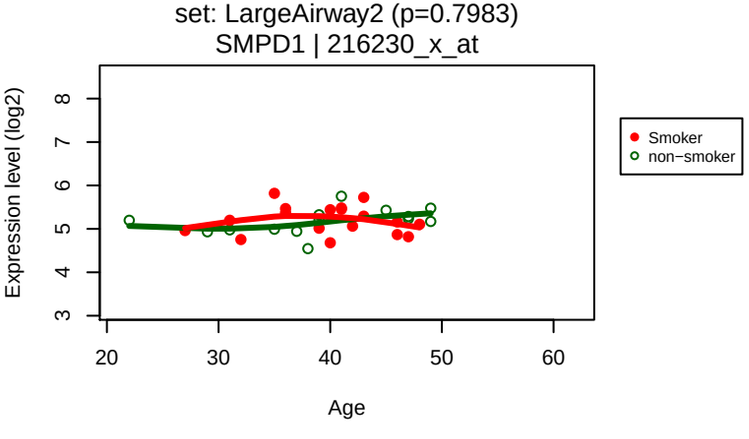

Gene expression ratio (smokers/nonsmokers)
gene: SMPD1 | 216230_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9781
p=0.5566
p=0.8748
p=0.4098
p=0.7983
p=0.0743
p=0.9851
p=0.3363
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

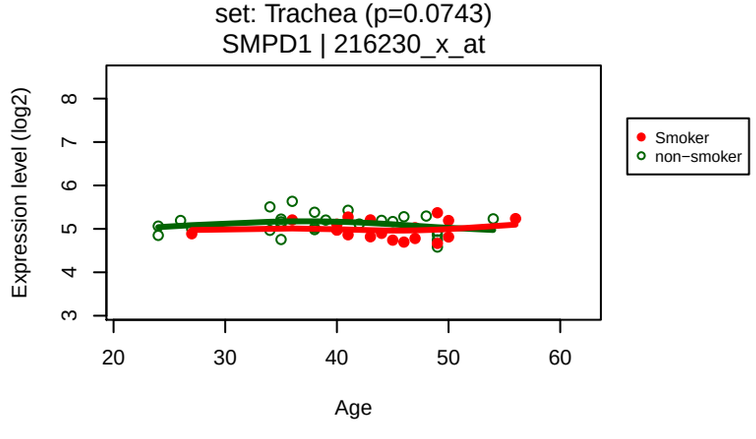

Gene expression ratio in Males
gene: SMPD1 | 216230_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3392
p=0.0476
p=0.496
p=0.1344
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMPD1 | 216230_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7611
p=0.0079
p=0.1229
p=0.2285
SmallAirway2
LargeAirway2
Trachea
Macrophage2

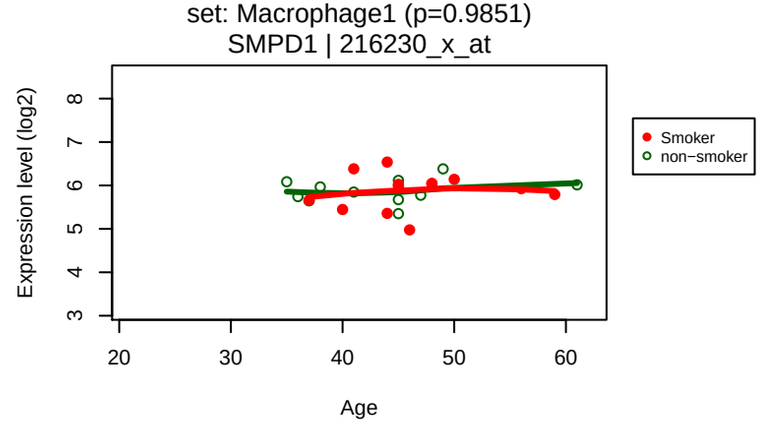

set: SmallAirway1 (p=0.9781)
SMPD1 | 216230_x_at
Expression level (log2)
Age
Smoker
non−smoker

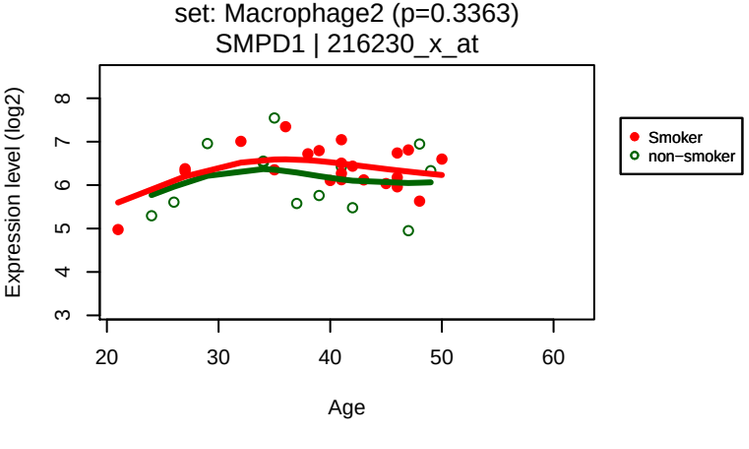

set: SmallAirway2 (p=0.5566)
SMPD1 | 216230_x_at
Expression level (log2)
Age
Smoker
non−smoker

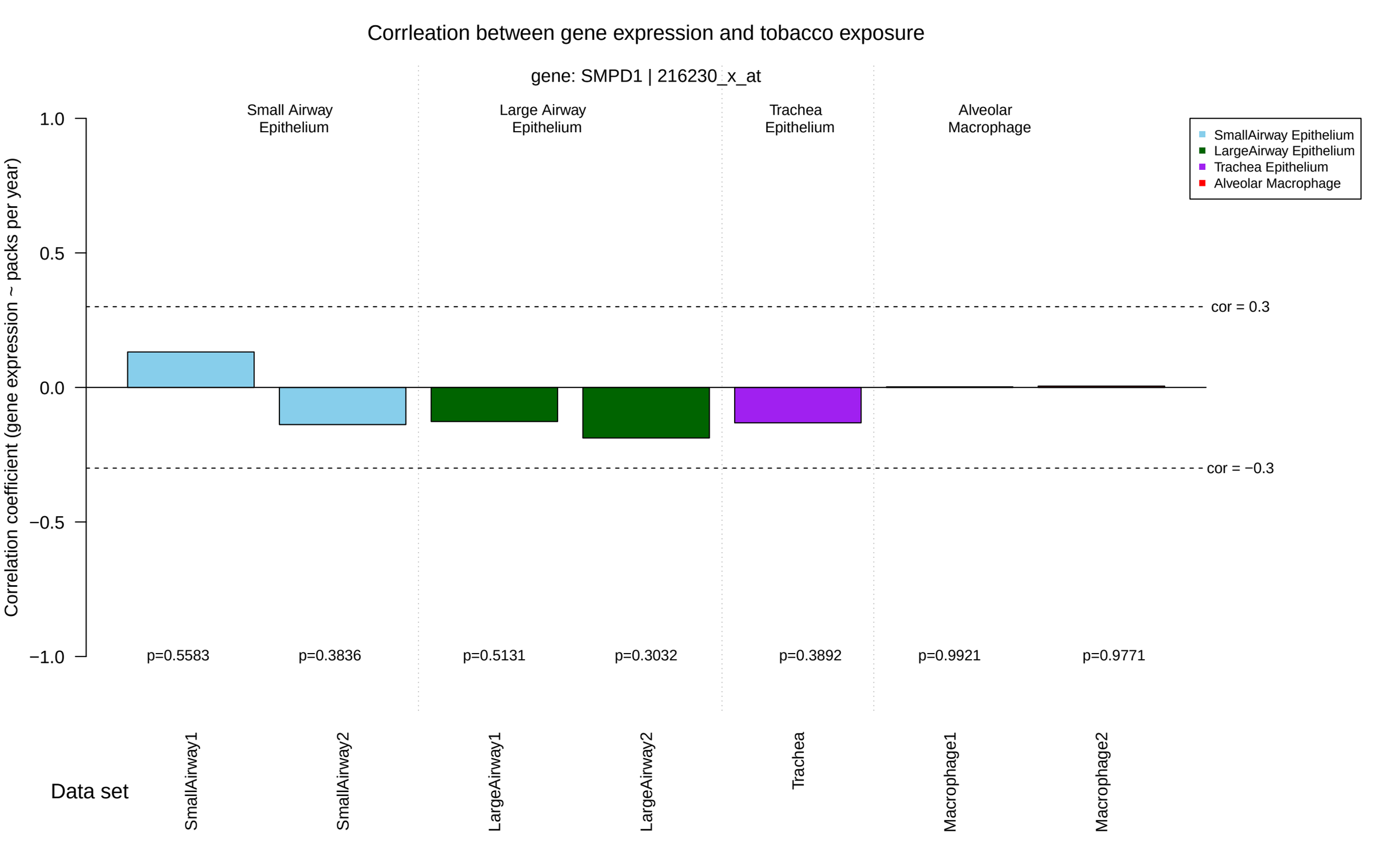

set: LargeAirway1 (p=0.4098)
SMPD1 | 216230_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7983)
SMPD1 | 216230_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0743)
SMPD1 | 216230_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9851)
SMPD1 | 216230_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3363)
SMPD1 | 216230_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMPD1 | 216230_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5583
p=0.3836
p=0.5131
p=0.3032
p=0.3892
p=0.9921
p=0.9771
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2